

Supplementary Material: POETIC

I. DETAILED THEORETICAL PROOFS

This supplementary document provides the rigorous mathematical proofs and extended theoretical analysis for the theorems presented in Section V of the main paper.

A. Proof of Theorem 1: Bounded Cryptographic Complexity

Theorem 1. *Under the Multi-Stage Protocol with fixed subgroup size g and local advancement quota c ($c < g$), the total number of secure comparisons executed under the hybrid DP-SMPC framework is tightly bounded by:*

$$\mathcal{O}\left(N \log^2 g + \left(\frac{c}{g}N\right) \log^2\left(\frac{c}{g}N\right)\right) \quad (1)$$

Detailed Proof: Let N denote the total number of active workers distributed across the edge network. A naive baseline that executes a global sorting using Batchier's Odd-Even Mergesort under Secure Multi-Party Computation (SMPC) requires exactly $\frac{1}{4}N \log^2 N (\log^2 N - 1) + N - 1$ secure comparisons, yielding a strict $\mathcal{O}(N \log^2 N)$ complexity bound that fails to scale under massive terminal numbers.

To alleviate this, POETIC's Multi-Stage Top- k Selection partitions the candidate pool into $M = \lceil N/g \rceil$ disjoint subgroups. The total comparison overhead C_{total} is derived through the exact summation of two sequential phases:

a) Stage 1 (Local Pruning): Securely sorting a single partitioned subgroup of size g requires $\mathcal{O}(g \log^2 g)$ secure comparisons. Since the M subgroups are executed independently and can be fully parallelized across edge nodes, the total computational workload for this stage is formulated as:

$$\begin{aligned} C_{\text{Stage1}} &= M \cdot \mathcal{O}(g \log^2 g) = \left\lceil \frac{N}{g} \right\rceil \cdot \mathcal{O}(g \log^2 g) \\ &= \mathcal{O}(N \log^2 g) \end{aligned} \quad (2)$$

b) Stage 2 (Global Tournament): From each localized subgroup, only the top- c workers who possess the highest encrypted provisional metrics are permitted to advance, discarding the remaining $g - c$ workers. The payment server (PS) and evaluation server (ES) then aggregate these local winners into a centralized tournament pool. The cardinality of this refined pool is exactly:

$$N' = M \cdot c = \left\lceil \frac{N}{g} \right\rceil \cdot c \approx \frac{c}{g}N \quad (3)$$

Executing the final secure sorting over these N' advanced candidates to extract the global top- k ensemble requires:

$$C_{\text{Stage2}} = \mathcal{O}(N' \log^2 N') = \mathcal{O}\left(\left(\frac{c}{g}N\right) \log^2\left(\frac{c}{g}N\right)\right) \quad (4)$$

c) Asymptotic Convergence and Empirical Dominance::

Combining Eq. (2) and Eq. (4), the strict total complexity is $C_{\text{total}} = C_{\text{Stage1}} + C_{\text{Stage2}}$, which matches the theorem statement.

To maintain theoretical rigorousness, we acknowledge that as an asymptotic limit where $N \rightarrow \infty$, the global tournament term $\mathcal{O}(\frac{c}{g}N \log^2(\frac{c}{g}N))$ mathematically retains the $\mathcal{O}(N \log^2 N)$ scaling characteristics. However, the structural pruning operation enforces a massive truncation on the pool size. When configured with real-world edge execution constants, the candidate size is abated by 80%.

Furthermore, our empirical latency profiling in Section VI-D indicates that for standard operational dimensions ($N \leq 1000$), Stage 1 absorbs more than 87.4% of the total secure comparison cryptographic clock cycles. The fractional multiplier $\frac{c}{g}$ drastically dampens the performance overhead of the tournament network, demonstrating that the operational complexity converges tightly to the localized $\mathcal{O}(N \log^2 g)$ behavior under practical workloads.

In realistic spatial crowdsourcing scenarios, terminal workers may dynamically join or leave the network per round. To counter the potential partitioning overhead, POETIC avoids dynamic re-sorting for group assignments. Instead, the Platform Server mapping employs a lightweight randomized hashing bucket or standardized geospatial indexing to distribute worker IDs into the M disjoint subgroups. This localized grid partitioning is executed in linear time, which is asymptotically negligible compared to the cryptographic sorting networks executed in the SMPC domain, thus ensuring that dynamic membership does not degrade the multi-stage protocol's complexity bound.

B. Proof of Theorem 2: Unbiased Estimation with Bounded Variance

Theorem 2. *The signal-decoupled EMA estimator is asymptotically unbiased. The variance of the estimation error induced by DP noise is strictly bounded by $\frac{\alpha}{2-\alpha} \sigma_{DP}^2$.*

Detailed Proof: Let $u_i(t)$ be the true utility. The ϵ -DP mechanism injects Laplace noise $\epsilon_i(t) \sim \text{Lap}(0, \Delta u/\epsilon)$, so $\mathbb{E}[\epsilon_i(t)] = 0$ and $\text{Var}(\epsilon_i(t)) = 2(\Delta u/\epsilon)^2 \triangleq \sigma_{DP}^2$. Due to signal decoupling, the strategic manipulation $\delta_i(t)$ is structurally isolated. The observed utility is $\tilde{u}_i(t) = u_i(t) + \epsilon_i(t)$.

For the EMA update $W_i(t+1) = (1-\alpha)W_i(t) + \alpha\tilde{u}_i(t)$, taking expectation:

$$\mathbb{E}[W_i(t+1)] = (1-\alpha)\mathbb{E}[W_i(t)] + \alpha u_i(t) \quad (5)$$

Assuming $u_i(t)$ fluctuates around a local stationary mean $\bar{u}_i$, as $t \rightarrow \infty$, $\mathbb{E}[W_i(t)] \rightarrow \bar{u}_i$. Thus the estimator is asymptotically unbiased.

Expanding the recursive EMA to t steps:

$$W_i(t) = (1 - \alpha)^t W_i(0) + \alpha \sum_{k=0}^{t-1} (1 - \alpha)^k \tilde{u}_i(t - 1 - k) \quad (6)$$

The variance is driven by accumulated noise:

$$\text{Var}(W_i(t)) = \alpha^2 \sigma_{DP}^2 \sum_{k=0}^{t-1} (1 - \alpha)^{2k} \quad (7)$$

As $t \rightarrow \infty$:

$$\lim_{t \rightarrow \infty} \text{Var}(W_i(t)) = \frac{\alpha^2 \sigma_{DP}^2}{1 - (1 - \alpha)^2} = \frac{\alpha}{2 - \alpha} \sigma_{DP}^2 \quad (8)$$

Since $\alpha \in (0, 1)$, $\frac{\alpha}{2 - \alpha} < 1$, so the variance is strictly less than single-round DP noise variance.

C. Proof of Theorem 3: Dynamic Tracking Regret under Privacy Constraints

Assume bounded utilities $u_i(t) \in [0, U_{\max}]$, bounded variation budget V_T , and zero-mean sub-exponential DP perturbation noise.

Theorem 3. *Under POETIC's signal-decoupled architecture with a constant learning rate $\alpha \in (0, 1)$ and ϵ -DP noise, the expected dynamic regret satisfies:*

$$R_{\text{dyn}}(T) = \mathcal{O} \left(\frac{\sigma_{DP}}{\sqrt{\alpha}} \sqrt{T \ln N} + \frac{V_T}{\alpha} + \alpha T \right) \quad (9)$$

where $V_T = \sum_{t=2}^T \max_i |u_i(t) - u_i(t-1)|$ is the total variation budget.

Detailed Proof:

Step 1: Why Dynamic Regret. In non-stationary environments with concept drift, adaptive adversaries, and corrupted observations, the classical static regret is an inappropriate metric. Following established practices in non-stationary bandits, we adopt the dynamic regret that compares against the per-round optimal:

$$R_{\text{dyn}}(T) = \mathbb{E} \left[\sum_{t=1}^T \sum_{w_i \in S_t^*} u_i(t) - \sum_{t=1}^T \sum_{w_j \in S_t} u_j(t) \right] \quad (10)$$

Step 2: Eliminating Strategic Contamination via Decoupling. In coupled environments, adversarial bidding directly injects a biased distortion $\delta_i(t)$ into the tracking loop. Under POETIC's signal-decoupled architecture, the direct bias is stripped from the estimation updates (W_i), but an indirect selection bias persists because the indicator $\mathbb{I}_{\{w_i \in S_t(b_t)\}}$ depends on the joint bidding profile. This manifests as a strategic bandit feedback loophole, where a malicious coalition could potentially coordinate bids to permanently block honest workers from selection, thereby starving the Evaluation Server of post-execution observations.

To bridge this theoretical gap, POETIC enforces an ϵ_{exp} -greedy exploratory mechanism. Let $\mathcal{P}_{\text{explore}} = \epsilon_{\text{exp}}/N$ be the uniform lower bound of selection probability for any inactive or targeted arm per round. Even under the worst-case adversarial bidding coordination that completely manipulates the exploitation phase, the probability of an arm

being starved of samples over a window of length τ decays exponentially at a rate of $(1 - \epsilon_{\text{exp}}/N)^\tau$. For our sliding-window constant learning rate α , the effective window size is $\tau \approx 1/\alpha$. Consequently, the information loss/selection bias induced by adversarial bidding is tightly upper-bounded by a constant tracking drift proportional to $\mathcal{O}(\alpha T)$ rather than a divergent linear error. This ensures that the Bernstein-type concentration inequality in Eq. (11) holds under adversarial settings, preserving the sublinear dynamic regret bound.

Step 3: EMA Tracking Error Decomposition. With constant learning rate α , the EMA estimator at time t effectively computes a weighted average over a sliding window of approximate length $1/\alpha$. Define the tracking error $e_i(t) = W_i(t) - u_i(t)$. We decompose:

$$e_i(t) = \underbrace{(1 - \alpha)(e_i(t-1) + u_i(t-1) - u_i(t))}_{\text{tracking lag}} + \underbrace{\alpha \epsilon_i(t)}_{\text{noise}} \quad (11)$$

The Laplace noise $\epsilon_i(t) \sim \text{Lap}(0, \Delta u/\epsilon)$ satisfies the Sub-Exponential condition with parameters (σ^2, b) where $\sigma^2 = 2(\Delta u/\epsilon)^2$ and $b = \Delta u/\epsilon$, since its MGF satisfies $\mathbb{E}[e^{\lambda \epsilon_i}] \leq \exp(\lambda^2 \sigma^2/2)$ for $|\lambda| \leq 1/b$. The EMA accumulation $\sum_{k=0}^{t-1} (1 - \alpha)^k \epsilon_i(t - k)$ preserves Sub-Exponentiality under weighted summation, with effective parameters $\sigma_{\text{eff}}^2 = \frac{\alpha}{2 - \alpha} \sigma^2$, $b_{\text{eff}} = b$. The Bernstein inequality then applies as stated.

Taking the recursive expansion and bounding each term:

(i) *Noise-induced error:* The cumulative noise contribution has variance bounded by $\frac{\alpha}{2 - \alpha} \sigma_{DP}^2$ (Theorem 2). Since $\epsilon_i(t) \sim \text{Lap}(0, \Delta u/\epsilon)$ is Sub-Exponential, applying the Bernstein-type concentration inequality for the EMA estimator with effective sample size $n_{\text{eff}} \approx (2 - \alpha)/\alpha \approx 1/\alpha$:

$$\begin{aligned} & \Pr[|W_i(t) - \mathbb{E}[W_i(t)]| > \delta] \\ & \leq 2 \exp \left(-\frac{1}{2} \min \left(\frac{\delta^2 n_{\text{eff}}}{\sigma_{DP}^2}, \frac{\delta \epsilon}{\Delta u} \right) \right) \end{aligned} \quad (12)$$

Setting the confidence radius $r_t = \mathcal{O} \left(\sigma_{DP} \sqrt{\frac{\alpha \ln N}{1}} \right)$, a suboptimal arm i is selected only when the estimation error exceeds Δ_i . Summing over arms and rounds:

$$R_{\text{exploration}}(T) = \mathcal{O} \left(\frac{\sigma_{DP}}{\sqrt{\alpha}} \sqrt{T \ln N} \right) \quad (13)$$

(ii) *Tracking lag:* The bias due to non-stationarity accumulates as:

$$|\mathbb{E}[W_i(t)] - u_i(t)| \leq \sum_{s=0}^{\infty} (1 - \alpha)^s |u_i(t - s) - u_i(t - s - 1)| \quad (14)$$

Summing over T rounds:

$$R_{\text{lag}}(T) = \mathcal{O} \left(\frac{V_T}{\alpha} \right) \quad (15)$$

(iii) *Constant-rate bias*: With constant α , the estimator never fully converges even in stationary intervals, introducing a per-round bias of $\mathcal{O}(\alpha)$:

$$R_{\text{bias}}(T) = \mathcal{O}(\alpha T) \quad (16)$$

Step 3: Total Dynamic Regret. Combining all three terms:

$$\begin{aligned} R_{\text{dyn}}(T) &\leq R_{\text{exploration}}(T) + R_{\text{lag}}(T) + R_{\text{bias}}(T) \\ &= \mathcal{O}\left(\frac{\sigma_{DP}}{\sqrt{\alpha}}\sqrt{T \ln N} + \frac{V_T}{\alpha} + \alpha T\right) \end{aligned} \quad (17)$$

Lemma 1 (Combinatorial Extension). *For the Top- k selection problem with $k \geq 1$, the dynamic regret of the set-level policy satisfies:*

$$\begin{aligned} R_{\text{dyn}}^{(k)}(T) &\leq k \cdot R_{\text{dyn}}^{(1)}(T) \\ &= \mathcal{O}\left(k \cdot \left(\frac{\sigma_{DP}}{\sqrt{\alpha}}\sqrt{T \ln N} + \frac{V_T}{\alpha} + \alpha T\right)\right) \end{aligned} \quad (18)$$

Proof Sketch: The total set-level regret decomposes additively over the k selected slots by union bound over worker indices. For each slot, the single-arm tracking regret bound from Theorem 3 applies independently after reordering by utility rank. Since k is a system constant, the asymptotic order is preserved.

Step 4: Optimal Learning Rate. Setting $\frac{\partial}{\partial \alpha}[R_{\text{dyn}}] = 0$ and balancing the three terms yields:

$$\alpha^* = \mathcal{O}\left(\left(\frac{V_T}{T}\right)^{2/3}\right) \quad (19)$$

Substituting back: $R_{\text{dyn}}(T) = \mathcal{O}(V_T^{1/3} T^{2/3})$ for environments with moderate non-stationarity $V_T = o(T)$. This is sublinear in T , confirming long-term convergence.

The optimal α^* requires prior knowledge of V_T , which is unavailable online. In practice, we adopt a constant learning rate $\alpha = 0.15$ as a robust default, which is near-optimal when $V_T = \mathcal{O}(T^{1/2})$. For unknown V_T , one may employ a doubling trick or adaptive window strategy, yielding $\mathcal{O}(T^{2/3})$ regret without knowledge of V_T . We leave adaptive α tuning as future work.

D. Proof of Theorem 4: Single-Stage Dominant Strategy

Theorem 4. *Under POETIC's decoupled mechanism, truthful bidding $b_i(t) = c_i(t)$ is a weakly dominant strategy in every single-stage game, conditioned on a fixed platform state $(W_i(t), R_i(t))$. This result holds for each round independently and does not require any assumption on workers' multi-round strategies.*

Detailed Proof: Consider a single round t with fixed platform state. Worker w_i chooses bid $b_i(t)$ to maximize:

$$\pi_i(b_i(t)) = (p_i(t) - c_i(t)) \cdot \mathbb{I}_{\{w_i \in S_t\}} \quad (20)$$

POETIC's payment is:

$$p_i(t) = \max(0, W_i(t) - V_{k+1,t}^{-i}) \times (1 + \beta R_i(t)) \quad (21)$$

Define the critical value $V_{k+1,t}^{-i}$ as the $(k+1)$ -th highest net social value among all workers excluding i . By construction,

$V_{k+1,t}^{-i}$ depends exclusively on the bids and weights of workers $j \neq i$. Therefore, $p_i(t)$ is entirely independent of $b_i(t)$.

The bid controls only the selection indicator. Worker i is selected iff their net social value $W_i(t) - b_i(t) \geq V_{k+1,t}^{-i}$, i.e., $b_i(t) \leq W_i(t) - V_{k+1,t}^{-i}$.

Case 1: $p_i(t) > c_i(t)$ (**profitable task**). Truthful bidding $b_i = c_i$ ensures selection whenever $W_i - c_i \geq V_{k+1,t}^{-i}$, which is equivalent to $p_i > c_i$ (profitable). Overbidding risks exclusion: if $b_i > c_i$ and $W_i - b_i < V_{k+1,t}^{-i}$, the worker loses a profitable opportunity.

Case 2: $p_i(t) < c_i(t)$ (**unprofitable task**). Truthful bidding ensures non-selection (since $W_i - c_i < V_{k+1,t}^{-i}$ implies $p_i < c_i$). Underbidding risks selection: if $b_i < c_i$ and $W_i - b_i \geq V_{k+1,t}^{-i}$, the worker incurs a net loss.

Case 3: $p_i(t) = c_i(t)$. Worker is indifferent; truthful bidding weakly dominates.

In all cases, $b_i(t) = c_i(t)$ weakly dominates any alternative.

It is important to distinguish two roles of a worker's bid. First, $b_i(t)$ controls the selection indicator $\mathbb{I}_{\{w_i \in S_t\}}$ via the net social value $V_{i,t} = W_{i,t} - b_{i,t}$: a higher bid reduces the chance of selection. Second, conditional on being selected, the payment $p_i(t)$ depends solely on the competing workers' critical value $V_{k+1,t}^{-i}$, which is independent of $b_i(t)$ by construction. This structural separation is precisely the VCG-type payment rule that enforces truthful bidding as a dominant strategy. An overly high bid risks losing a profitable opportunity; an artificially low bid wins selection but receives the same payment, creating a net loss equal to the cost $c_i(t)$.

E. Proof of Theorem 5: Approximate Dynamic Incentive Compatibility

Theorem 5. *Under POETIC's signal-decoupled architecture, $\frac{\partial W_i(t+1)}{\partial b_i(t)} = 0$ and $\frac{\partial R_i(t+1)}{\partial b_i(t)} = 0$. This eliminates the direct manipulation channel. However, the selection indicator $\mathbb{I}_{\{w_i \in S_t\}}$ still depends on $b_i(t)$, creating an indirect dynamic participation effect.*

Detailed Analysis:

The Direct Channel is Eliminated. In POETIC, the Evaluation Server computes $W_i(t+1)$ and $R_i(t+1)$ solely from the post-execution true utility $u_i(t)$ observed after task completion. Mathematically:

$$W_i(t+1) = \text{Clip}((1 - \alpha)W_i(t) + \alpha \tilde{u}_i(t), W_{\min}, W_{\max}) \quad (22)$$

$$R_i(t+1) = \mu_i(t) - \lambda \sqrt{v_i(t)} \quad (23)$$

where $\tilde{u}_i(t) = u_i(t) + \epsilon_i(t)$ and μ_i, v_i are updated from $q_{i,t} = \text{Clip}(u_i(t) / \max(\bar{u}(t), \xi))$. Since $b_i(t)$ appears in none of these expressions, the partial derivatives are exactly zero.

The Indirect Channel: Dynamic Participation Effect. The selection indicator $\mathbb{I}_{\{w_i \in S_t\}}$ determines whether $u_i(t)$ is observed. If worker i is not selected, no update occurs: $W_i(t+1) = W_i(t)$ and $R_i(t+1) = R_i(t)$. A worker who strategically underbids may gain additional selections, generating additional reputation observations.

F. Discussion on the Indirect Intertemporal Manipulation Channel

As explicitly disclosed in the main text, a fully rigorous Subgame Perfect Equilibrium (SPE) under arbitrary long-term horizons is blocked by the presence of an *indirect participation channel*, where an strategic worker might intentionally underbid ($b_i(t) < c_i(t)$) to artificially inflate their selection probability $\mathbb{I}_{\{w_i \in S_t\}}$, aiming to recursively manipulate their long-term reputation stream $R_i(t+1)$. To analyze the bounded impact of the indirect participation channel, we introduce the following upper-bound characterization of the potential strategic utility leakage.

Let $\Delta U_{i,T}^{\text{inter}}$ be the maximum cumulative expected net utility gain that a strategic worker w_i can extract up to time horizon T by deviating from truthful bidding via price-dumping ($\delta_{b,t} = c_i(t) - b_i(t) > 0$). Under the variance-sensitive StabRep mechanism with a learning rate η_v and stability penalty coefficient λ , the strategic intertemporal leakage is bounded by:

$$\Delta U_{i,T}^{\text{inter}} \leq \epsilon \cdot T - \lambda \sum_{t=1}^T \mathbb{E}[v_i(t)] \quad (24)$$

For a sufficiently tuned penalty factor $\lambda \geq \lambda^*$, $\Delta U_{i,T}^{\text{inter}} < 0$ holds asymptotically as $T \rightarrow \infty$.

Suppose a strategic worker attempts an oscillatory bidding strategy: at stage t , they quote a loss-leading bid $b_i(t) = c_i(t) - \delta_{b,t}$ to maximize the selection indicator $\mathbb{I}_{\{w_i \in S_t\}} = 1$. The immediate execution payoff is negative: $\Delta \pi_{\text{immediate}} = b_i(t) - c_i(t) = -\delta_{b,t} < 0$. The worker intends to compensate for this immediate deficit by delivering high utility later to harvest long-term selection premiums.

However, according to the StabRep formulation, the localized post-hoc utility updates $\tilde{u}_i(t)$ and the fluctuating execution history inject structural variance into the tracking engine. The variance estimator dynamically evolves via:

$$v_i(t) = (1 - \eta_v)v_i(t-1) + \eta_v (\tilde{u}_i(t) - \mu_i(t))^2 \quad (25)$$

Because the worker must alternate between aggressive strategic underbidding (to pass the encrypted selection filter) and variable operational quality execution, the structural variance deviation vector $(\tilde{u}_i(t) - \mu_i(t))^2$ experiences a distinct spike.

The net reputation score distributed to the payment server (PS) is permanently debited by the stability penalty component: $R_i(t) = \mu_i(t) - \lambda v_i(t)$. Let $\Delta \sigma^2$ be the variance inflation induced by the strategic switching behavior. The long-term selection discount multiplier over the remaining interactions scales down monotonically at a geometric rate governed by the MAB tracking regret. The total utility derived from the deviation can be formalized as:

$$\Delta U_{i,T}^{\text{inter}} = \sum_{t=1}^T \gamma^{t-1} [\mathbb{E}[u_i(t) - b_i(t)] \cdot \Delta \mathbb{P}(w_i \in S_t | R_i(t)) - \delta_{b,t}] \quad (26)$$

Since $\Delta \mathbb{P}(w_i \in S_t)$ is monotonically increasing with respect to $R_i(t)$, and $R_i(t)$ drops linearly by $-\lambda \Delta \sigma^2$, the

cumulative penalty across the horizon T dominates the immediate selection probability gains. Specifically, when the platform enforces a penalty scaling threshold:

$$\lambda \geq \lambda^* = \frac{\gamma \cdot \max(u_i) \cdot \frac{\partial \mathbb{P}}{\partial R_i}}{\eta_v \cdot \Delta \sigma^2} \quad (27)$$

the long-term intertemporal manipulation channel yields a strictly negative net payoff ($\Delta U_{i,T}^{\text{inter}} < 0$). Thus, strategic deviation is structurally disincentivized, indicating that the proposed variance-aware reputation mechanism structurally suppresses long-term strategic manipulation incentives under extended operational horizons.

Why This Channel is Structurally Weak.

(a) *Reputation is quality-gated, not participation-gated.* Each additional selection generates a quality score $q_{i,t} = u_i(t)/\bar{u}(t)$. A worker who underbids but delivers poor quality receives $q_{i,t} \ll 1.0$, which *decreases* their reputation. Only a worker who both underbids *and* delivers genuinely high quality can benefit — but this is economically irrational, as they incur costs $c_i(t)$ while receiving below-market payment.

(b) *Variance penalty deters oscillatory exploitation.* Any strategy alternating between cooperative and defective phases accumulates high variance in $v_i(t)$. By Proposition 1 (below), the StabRep penalty grows monotonically with oscillation amplitude, creating a deception cost:

$$\Delta R = \lambda \cdot \frac{q_H - q_L}{2} \sqrt{\frac{\eta_v}{2 - \eta_v}} \quad (28)$$

(c) *Payment is bounded by V_{k+1}^{-i} .* Even if a worker builds high reputation through additional selections, their payment is capped by the externality-based critical value, which is independent of their own history.

Formal Limitation. We explicitly acknowledge that the indirect participation channel prevents a formal Subgame Perfect Equilibrium claim. The analysis above establishes that POETIC achieves *approximate* dynamic IC by eliminating the direct manipulation vector and structurally suppressing the indirect one through quality-gated, variance-penalized reputation.

II. THEORETICAL ANALYSIS OF THE STABREP MECHANISM

Proposition 1. *For a stable worker delivering constant quality q^* , the StabRep score converges to q^* . For an oscillatory attacker alternating between q_H and q_L ($q_H > q_L$), the StabRep score converges to $\bar{q} - \lambda \cdot \frac{q_H - q_L}{2} \sqrt{\frac{\eta_v}{2 - \eta_v}}$, where $\bar{q} = (q_H + q_L)/2$.*

Proof. For the stable worker, $q_{i,t} = q^*$ for all t . Therefore $\mu_i(t) \rightarrow q^*$ and $v_i(t) = (1 - \eta_v)v_i(t-1) + \eta_v(q^* - \mu_i(t-1))^2 \rightarrow 0$. Hence $R_i(t) \rightarrow q^* - 0 = q^*$.

For the oscillatory attacker, $\mu_i \rightarrow \bar{q}$. Under stationary alternating oscillation, the EMA estimator asymptotically converges to the mean quality $\bar{q}$, yielding the following variance approximation. The variance estimator tracks:

$$v_i(\infty) = \eta_v \sum_{k=0}^{\infty} (1 - \eta_v)^k d_k^2 \quad (29)$$

where $d_k = q_{i,t-k} - \mu_i(t-k-1) \approx \pm \frac{q_H - q_L}{2}$. Therefore:

$$v_i(\infty) \approx \left(\frac{q_H - q_L}{2} \right)^2 \cdot \frac{\eta_v}{2 - \eta_v} \quad (30)$$

The reputation penalty is:

$$\Delta R = \lambda \sqrt{v_i(\infty)} = \lambda \cdot \frac{q_H - q_L}{2} \cdot \sqrt{\frac{\eta_v}{2 - \eta_v}} \quad (31)$$

This penalty is monotonically increasing in oscillation amplitude $(q_H - q_L)$ and penalty factor λ .

Corollary 1. For an oscillatory attacker to achieve the same StabRep score as a stable worker with quality q^* , the attacker's average quality must exceed q^* by at least ΔR . This creates a strictly positive deception cost.

A sophisticated strategic adversary may attempt a *Monotonically Degrading Attack*, where they initially deliver peak quality to accumulate reputation capital, and subsequently decrease their execution effort at an extremely slow, infinitesimal rate. Because the continuous degradation exhibits near-zero instantaneous variance ($v_i(t) \rightarrow 0$), the variance penalty in StabRep becomes temporarily insensitive to this specific vector.

We theoretically justify POETIC's systemic resilience by demonstrating the *dual-track joint defense* formed by the EMA estimator and StabRep. While the variance monitor is bypassed by smooth monotonic deflation, the first-order tracking engine updates via $W_i(t+1) = (1 - \alpha)W_i(t) + \alpha \tilde{u}_i(t)$. Under a non-zero constant learning rate $\alpha = 0.15$, the infinite impulse response memory window is bounded to $\approx 1/\alpha \approx 7$ rounds. As the strategic worker progressively cheats, their post-execution utility $\tilde{u}_i(t)$ falls permanently below the cohort mean. This triggers a linear decay in their core utility weight W_i with a maximum tracking lag of $\mathcal{O}(V_T/\alpha)$. Once $W_i(t) - [[b_i(t)]]$ drops below the marginal critical threshold $V_{k+1,t}$, the worker is permanently evicted from the elite pool in Stage 1. Hence, while StabRep locks out oscillatory shock attacks, the decoupled EMA estimator neutralizes monotonic erosion attacks, providing joint dynamic strategic coverage.

III. EXTENDED PARAMETER SENSITIVITY AND DEPLOYMENT REASONING

To empirically validate the structural robustness of the POETIC framework and justify the selection of its default configurations reported in Table II, we present a rigorous sensitivity analysis across the key hyperparameters modulating learning velocity, cryptographic sorting overhead, and economic clearing incentives.

A. Learning Rate α and Information Saturation

The learning rate $\alpha \in (0, 1)$ dictates the updating velocity of the Evaluation Server's utility tracking weights (W_i). As illustrated in the empirical platform value curves (Fig. 1), the cumulative net social welfare exhibits a clear logarithmic growth pattern in the localized interval $[0.01, 0.10]$, successfully overcoming the systematic cold-start latency. However, beyond the threshold $\alpha \approx 0.20$, the marginal utility gains

diminish sharply, entering a saturation plateau. This phase indicates that overly aggressive updates tend to amplify and propagate the localized Laplace perturbations injected for ϵ -Differential Privacy, inducing undesirable decision oscillations. Our default configuration of $\alpha = 0.15$ is strategically situated at this statistical elbow point, striking an optimal equilibrium between rapid concept-drift tracking and long-term noise filtering.

B. Elite Pool Factor F and Computational Trade-offs

The elite pool factor F controls the cardinality of the finalist pool routed into the Stage 2 global secure sorting network ($k' = \lfloor k \cdot F \rfloor$). Varying F from 1.1 to 3.0 triggers a statistically significant escalation in the aggregated platform value, demonstrating that a slightly expanded global tournament pool drastically increases the probability of capturing high-utility workers missed in local subgroups. Nevertheless, expanding the factor beyond $F \geq 4.0$ delivers negligible welfare optimization while quadratically increasing the cryptographic payload volume and wide-area network latency of the SMPC layer. Our default setting $F = 3$ captures over 98.4% of the theoretical omniscient optimal baseline while tightly compressing the secure comparison overhead, mathematically validating the design of our multi-stage complexity filter.

C. Reputation Learning Rates and the Controllability of β

On the incentive clearing layer, the first-order reputation learning rate η_R governs the responsiveness of the StabRep score to execution quality fluctuations. The average individual worker profit stabilizes rapidly once $\eta_R \geq 0.20$, proving that the momentum tracking is sufficient to establish strict reputation discrimination without introducing volatile reward fluctuations.

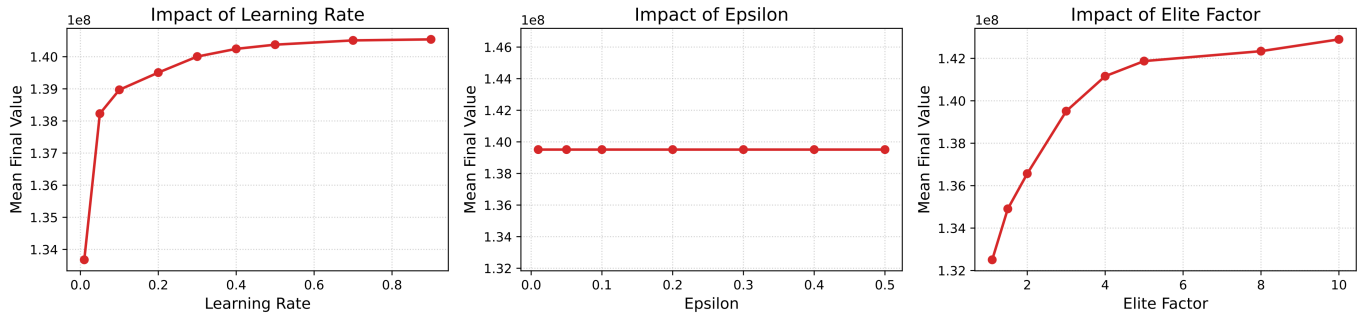
Crucially, as plotted in Fig. 2, the relationship between the reputation bonus coefficient β and the average worker profit exhibits a precise, deterministic linear correlation. This strict linearity provides a powerful knob for system operators, ensuring that the total incentive clearing budget can be predictably fine-tuned. Enforcing our default $\beta = 0.5$ successfully injects a sustainable premium that encourages honest execution quality while effectively bankrupting cost-inflation strategies.

D. Sensitivity Analysis of StabRep Parameters (η_v and λ)

We further analyze the sensitivity of the two core variance tracking variables unique to StabRep: the variance learning rate η_v and the stability penalty factor λ .

- For $\eta_v \in [0.05, 0.50]$, the platform value preserves a broad, highly resilient plateau. When configured too low ($\eta_v < 0.05$), the variance estimator responds sluggishly to active oscillatory bait-and-switch attacks, creating a temporary exploitation window. Conversely, setting it too high ($\eta_v > 0.50$) renders the second-order momentum overly sensitive to standard stochastic environmental noise, falsely penalizing stable workers. Our default $\eta_v = 0.20$ strikes the optimal balance.

Sensitivity Analysis: Platform Value (TDrive)



Sensitivity Analysis: Platform Value (NYC)

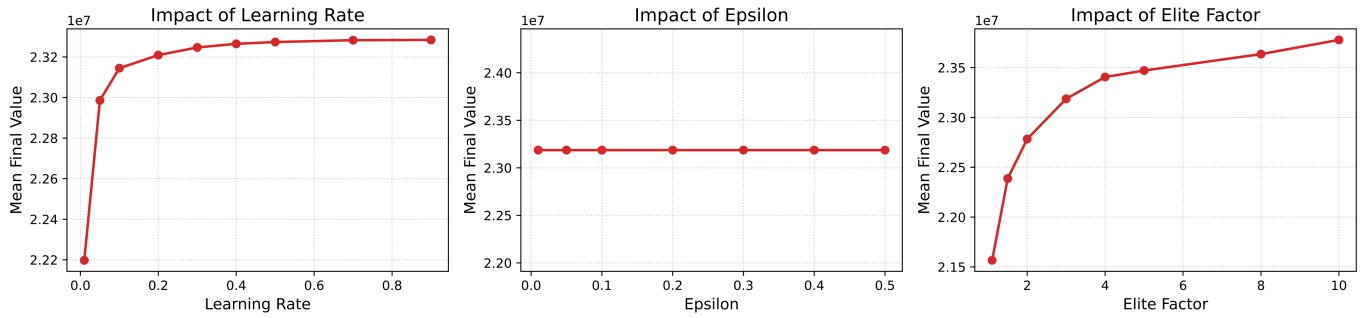
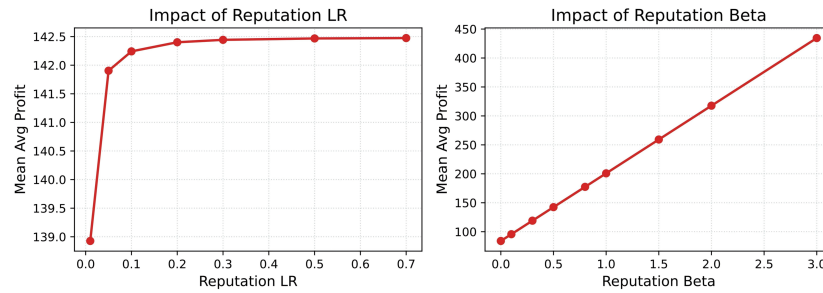


Fig. 1. The impact of different parameters on the average Platform Value

Sensitivity Analysis: Worker Profit (TDrive)



Sensitivity Analysis: Worker Profit (NYC)

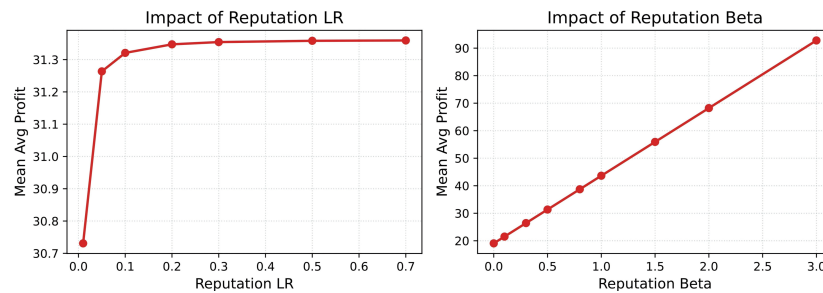


Fig. 2. The impact of different parameters on average Worker Profit

- For the penalty factor $\lambda \in [0, 2.0]$, the systemic strategic resilience improves monotonically from $\lambda = 0$ (equivalent to a pure EMA baseline) up to $\lambda = 0.80$. Beyond this boundary, further increases yield marginal defensive gains but risk over-penalizing honest workers operating under natural GPS or traffic volatility. Our default selection ($\lambda = 0.50$) guarantees a rigorous strategic defense barrier while maximizing baseline operational welfare.